



Chapter	Title	Total periods	Total time	Date
Ch 09	The Baudhayana-Pythagoras Theorem	10	480 min	24 May 2026

C No.	Targeted competencies
C-4.2	Learns the Baudhayana-Pythagoras theorem on the lengths of the sides of a right-angled triangle, and discovers a geometric proof using areas of squares erected on the sides of the triangle, and other related geometric constructions from the Sulba-Sutras
C-4.1	Discovers, understands, and uses formulae to determine the area of a square, triangle, parallelogram, and trapezium and develops strategies to find the areas of composite 2D shapes
C-3.2	Outlines the properties of lines, angles, triangles, quadrilaterals, and polygons and applies them to solve related problems
C-1.4	Explores and understands sets of numbers, such as whole numbers, fractions, integers, rational numbers, and real numbers, and their properties, and visualises them on the number line
C-9.2	Knows and appreciates the contributions of specific Indian mathematicians (such as Baudhayana, Pingala, Aryabhata, Brahmagupta, Virahanka, Bhaskara, and Ramanujan)

Period 1	40 min	Why Does the Diagonal Square Double the Area?
Material	Scissors, Square paper sheets (two per student pair), Ruler, Pencil	Pedagogical approach: Inductive
0-8	Pose Baudhayana's opening question: how do you build a square with exactly double the area of a given square? Students sketch a guess individually, then share with a neighbour to surface the common error of simply doubling the side.	
8-20	Walk through the geometric reasoning using the diagonal construction and auxiliary east-west/north-south lines. Use Math Talk, section 2.1, p.34 ("Why should the extension of the vertical and horizontal sides....") to prompt students to articulate why the four triangles in the original square equal the four in the new one.	
20-40	Students carry out Activity 1 (Doubling a Square Using Paper), p.35 ("Cut two identical squares of paper; cut Square 1 along diagonals....") in pairs. They assemble the pieces and verify the area relationship, then discuss in groups why the tiling works.	
Teacher Notes	This is the first period of the chapter, so spend a moment contextualising Baudhayana's Sulba-Sutra before the activity. Watch for students who think doubling the side doubles the area - the paper-cutting step should correct this viscerally. Keep group discussion brief so the assembly is completed within the period.	

Period 2	40 min	Folding Inward to Find the Half-Area Square
Material	Square paper sheets (one per student), Ruler, Pencil, Blackboard/whiteboard for proof steps	Pedagogical approach: Deductive
0-6	Briefly connect to last period: doubling used the diagonal; today the reverse - halving - uses midpoints. Ask students to predict what shape appears when all four midpoints of a square's sides are joined.	
6-20	Students perform Activity 2 (Halving a Square Using Paper), p.36 ("Cut out a square from paper; fold it inward such that crease lines pass through midpoints...."). They identify quadrilateral PQRS from the crease lines and record its apparent properties.	
20-35	Guide the class through the deductive proof: using east-west and north-south auxiliary lines, triangle congruence, and diagonal properties to establish that PQRS is a square and has exactly half the area. Students write each step in their own words.	
35-40	Exit check: each student states one geometric fact (in a sentence) that guarantees PQRS is a square. Collect responses verbally to gauge understanding before the next period.	
Teacher Notes	Students built intuition for the doubling construction last period; the halving argument is its mirror and should feel natural. Watch for confusion between showing all sides are equal (rhombus) versus also proving all angles are right angles (square) - both steps are needed. The diagonal-bisection argument is the key link to emphasise.	



Period 3	60 min	From $\sqrt{2}$ to Irrational: Reasoning About the Hypotenuse
Material	Square paper (for Q1 warm-up), Calculator (optional, for bounding checks), Blackboard, Rulers	Pedagogical approach: Inductive
0-8	Revisit the doubling result: the diagonal of a unit square has area 2 built on it. Use Figure it Out Q1, section 2.3, p.39 ("Two identical square papers are each cut diagonally....") as a quick hands-on warm-up to anchor $c^2 = 2$ for the unit isosceles right triangle.	
8-20	Lead the class inductively through successive bounding of $\sqrt{2}$: establish $1 < \sqrt{2} < 2$, then narrow to $1.4 < \sqrt{2} < 1.5$, then $1.41 < \sqrt{2} < 1.42$, recording each squeeze on the board. Students identify the pattern of bounding and predict the next interval.	
20-35	Use Try This, section 2.3, p.39 ("Can $\sqrt{2}$ be expressed as a fraction m/n") as a guided group investigation. Groups reason via last-digit analysis and prime factorisation arguments, then share findings; teacher consolidates the proof that $\sqrt{2}$ is irrational.	
35-48	Walk through Example 1, section 2.3, p.40 ("Find the hypotenuse of an isosceles right triangle whose equal sides have length 12....") and Example 2, section 2.3, p.40 ("If the hypotenuse of an isosceles right triangle is $\sqrt{72}$") on the board, emphasising how $c^2 = 2a^2$ drives both directions of the problem.	
48-60	Students work individually on Figure it Out Q2, section 2.3, p.39 ("Find the length of the hypotenuse and find bounds....") for all five parts. Teacher circulates and addresses bounding errors; assign Figure it Out Q3, section 2.3, p.40 as homework.	
Homework	Figure it Out Q3, section 2.3, p.40 — The hypotenuse of an isosceles right triangle is 10. Find its other two sidelengths. [Hint:...	
Teacher Notes	Students have now seen area-doubling geometrically; this period extends that into the formula $c^2 = 2a^2$ and the surprising irrationality of $\sqrt{2}$. Watch for students who try to simplify $\sqrt{288}$ to a rational - use the bounding method to show the value is genuinely between two integers. Example 2 is a good self-study pointer for students who finish early.	

Period 4	40 min	Applying the Hypotenuse Formula Across Varied Problems
Material	Exercise books, Blackboard, Ruler (for any sketch diagrams)	Pedagogical approach: Problem-solving
0-8	Review homework: discuss Figure it Out Q3, section 2.3, p.40 ("The hypotenuse of an isosceles right triangle is 10....") as a class. Highlight the hint about composing two triangles into a square, connecting back to the area reasoning of section 2.1.	
8-25	Students work in pairs on any remaining parts of Figure it Out Q2, section 2.3, p.39 ("Find the length of the hypotenuse and find bounds...."), focusing on producing clean decimal bounds. Pairs compare answers and resolve discrepancies through discussion.	
25-40	Whole-class debrief: select two pairs to present contrasting approaches (algebraic vs. bounding). Teacher closes by posing the bridge question - what happens when the two sides of a right triangle are not equal? - to motivate section 2.4.	
Teacher Notes	The previous period established the formula and irrationality reasoning; this period consolidates application before moving to the general theorem. A common error is using $c^2 = a^2 + a^2$ but then forgetting to take the square root - reinforce the two-step process. Students who struggled with bounding in period 3 should be paired with stronger peers.	

Period 5	50 min	Building the General Theorem from Two Different Squares
Material	Scissors, Square paper in two different sizes per pair, Ruler, Protractor, Pencil	Pedagogical approach: Discovery
0-8	Bridge from the isosceles case: show two different-sized squares on the board and ask students to guess the area of the square built on the diagonal of the right triangle with those side lengths. Record guesses without confirming them.	
8-28	Students carry out Activity 3 (Combining Two Squares Using Paper), p.45 ("Cut out and join two different-sized squares (sides a and b); make two cuts....") in pairs. They discover the rearrangement into a larger square and measure all sides to conjecture $a^2 + b^2 = c^2$.	
28-42	Lead the geometric proof: four congruent right-triangle copies placed around the hypotenuse square; use angle complementarity to prove all angles of the inner figure are right angles, confirming it is a square with area $a^2 + b^2$.	
42-50	Students attempt Figure it Out Q3, section 2.4, p.47 ("How would you construct a square whose area is triple....") as a discovery extension, sketching constructions and comparing with a neighbour.	



Teacher Notes	Students arrive with the special (isosceles) case well practised; the discovery activity now generalises it. The most common reasoning gap is proving all four angles of the central figure are 90° - the complementarity argument (two acute angles of a right triangle summing to 90°) needs to be made explicit. Worked example (Using Baudhayana's Theorem), section 2.4, p.47 with the 3-4-5 triangle is a good self-study reference for students who want a concrete numerical anchor.
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Period 6	50 min	Applying Baudhayana's Theorem to Find Missing Sides
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Material	Exercise books, Ruler, Blackboard	Pedagogical approach: Problem-solving
0-10	Quick review of the proof from last period; walk through Worked example (Using Baudhayana's Theorem), section 2.4, p.47 ("Make a right-angled triangle with shorter sides 3 cm and 4 cm....") together, emphasising the draw-then-verify workflow.	
10-22	Students individually solve Figure it Out Q1, section 2.4, p.47 ("If a right-angled triangle has shorter sides of lengths 5 cm and 12 cm....") and Figure it Out Q2, section 2.4, p.47 ("If a right-angled triangle has a short side of 8 cm and hypotenuse of 17 cm...."), drawing diagrams before applying algebra.	
22-42	Students work through Figure it Out Q4, section 2.4, p.47 ("Find the missing sidelength: (i) $a=5$, $b=7$; (ii)....") in small groups, noting which sub-parts require finding a leg versus the hypotenuse. Groups reconcile any discrepancies before sharing answers.	
42-50	Whole-class discussion of sub-part (iii) ($a=9$, $c=15$) as representative of the leg-finding case. Teacher reinforces rearranging $a^2 + b^2 = c^2$ to $b^2 = c^2 - a^2$ and closes with a preview of integer triples in section 2.5.	
Teacher Notes	Students proved the theorem last period; this period is pure application fluency. Watch for the common error of using $a^2 + b^2 = c^2$ when c is not the hypotenuse - part (iii) of Q4 specifically targets this. Encourage students to always label which side is the hypotenuse before writing any equation.	

Period 7	40 min	Generating Infinitely Many Integer Right Triangles
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Material	Exercise books, Blackboard, Number chart or multiplication table (optional)	Pedagogical approach: Inductive
0-8	Pose the question: are there right triangles with all integer sides besides 3-4-5? Students individually brainstorm and list candidates. Collect on the board to motivate a systematic approach.	
8-20	Use Math Talk, section 2.5, p.48 ("List down all the Baudhayana triples with numbers less than or equal to 20") as a structured group task. Groups record triples, identify which are multiples of others, and introduce the primitive vs. non-primitive distinction inductively.	
20-35	Introduce the identity $(n-1)^2 + (2n-1)^2 = n^2$ and walk through $n=5$ and $n=13$ together. Students then tackle Figure it Out Q1, section 2.5, p.50 ("Find 5 more Baudhayana triples using the odd-square method....") and begin Figure it Out Q2, section 2.5, p.50 ("Does the odd-square method yield non-primitive Baudhayana triples?....").	
35-40	Brief class share: confirm that the method always yields primitive triples and that scaling produces non-primitive ones. Assign Figure it Out Q3, section 2.5, p.50 as homework to investigate the method's completeness.	
Homework	Figure it Out Q3, section 2.5, p.50 — Are there primitive Baudhayana triples that cannot be obtained through the odd-square method? If yes,...	
Teacher Notes	Students have been applying the theorem numerically; this period shifts to algebraic reasoning about triples. Watch for students who confuse primitive triples (no common factor) with triples produced by the odd-square method - Q2 specifically targets this subtlety. The proof that (ka, kb, kc) is also a triple follows directly from squaring, and should be derived with the class rather than stated.	

Period 8	60 min	Fermat's Conjecture and Wiles's Proof: A 300-Year Story
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Material	Blackboard, Short printed timeline of Fermat's Last Theorem (optional handout)	Pedagogical approach: Discovery
0-10	Recap: students have shown infinitely many Baudhayana triples exist for $n=2$. Now pose the question: what if the exponent were 3 or higher? Students guess whether integer solutions might still exist for $x^n + y^n = z^n$ with $n > 2$.	



10-28	Present Fermat's 17th-century marginal note and his claim of a proof never written down. Students discuss in groups: why might a mathematician claim a proof without writing it? What would it take to verify or refute the claim over centuries?
28-45	Narrate the 300-year history of failed attempts by great mathematicians, leading to Andrew Wiles who set himself this goal as a 10-year-old in 1963 and completed the proof in 1994. Students reflect in writing: what does this story tell us about mathematical persistence?
45-60	Small-group discussion: connect Fermat's Last Theorem back to the chapter - why does the existence of infinitely many Baudhayana triples make Fermat's question natural? Groups share one-sentence summaries and teacher closes with the distinction between an open problem and a solved one.
Teacher Notes	Section 2.6 has no enumerated exercises but is rich in mathematical culture; the 60-minute period allows genuine discussion rather than just narration. Watch for students who conflate 'unproven for a long time' with 'impossible to prove' - the Wiles story directly addresses this. Encourage students to look up other open problems (Goldbach's conjecture, twin primes) as a voluntary extension.

Period 9	60 min	Lotus Stems, Rhombuses, and Grid Squares: Real Applications	
Material	Grid/dot paper, Ruler, Compass (optional for Q7 construction), Exercise books	Pedagogical approach: Problem-solving	
0-10	Open with the Bhaskaracharya Lilavati lotus-stem problem as a narrative hook, solving it together as a class to model how real-world situations translate into right-triangle algebra. Emphasise the setup - identifying which segment is the hypotenuse - before any computation.		
10-22	Students work individually on Figure it Out Q1, section 2.7, p.52 ("Find the diagonal of a square with sidelength 5 cm") and Figure it Out Q2, section 2.7, p.52 ("Find the missing sidelengths in the following right triangles...."). Teacher circulates to address errors in identifying legs versus hypotenuse.		
22-38	Pairs tackle Figure it Out Q3, section 2.7, p.53 ("Find the sidelength of a rhombus whose diagonals are of length 24 units and 70 units") and Figure it Out Q4, section 2.7, p.53 ("Is the hypotenuse the longest side...."), requiring both computation and written justification.		
38-52	Small groups explore Figure it Out Q6, section 2.7, p.53 ("Give 5 examples of rectangles whose sidelengths and diagonals are all integers") and Figure it Out Q8, section 2.7, p.53 ("Using dots of a grid as vertices, create squares of various areas...."), connecting Baudhayana triples to integer rectangles and grid geometry.		
52-60	Whole-class debrief of Q8 part (ii): what integer areas are achievable? Students conjecture and discuss. Assign Figure it Out Q5 and Q7 as homework and preview the final period's consolidation questions.		
Homework	Figure it Out Q5, section 2.7, p.53 — True or False — Every Baudh■yana triple is either a primitive triple or a scaled... Figure it Out Q7, section 2.7, p.53 — Construct a square whose area is equal to the difference of the areas of squares...		
Teacher Notes	The Lilavati problem is an elegant entry point - students often find it surprising that a poetic riddle reduces to a clean right-triangle setup. For Q3 (rhombus), watch for students who do not recognise that diagonals of a rhombus bisect each other at right angles; this property should be recalled rather than proved here. Q8 is rich enough for gifted students to explore beyond the period.		

Period 10	40 min	Consolidating Applications: Equilateral Triangles and Integer Rectangles	
Material	Ruler, Compass, Grid paper, Exercise books, Blackboard	Pedagogical approach: Problem-solving	
0-8	Discuss Figure it Out Q5, section 2.7, p.53 ("True or False - Every Baudhayana triple is either a primitive triple or a scaled version of a primitive triple") as homework review, connecting back to section 2.5's algebraic proof. Confirm the true/false verdict with a class vote and then the justification.		
8-25	Students work individually on Figure it Out Q9, section 2.7, p.54 ("Find the area of an equilateral triangle with sidelength 6 units...."), guided by the hint to first establish the altitude via bisection. Teacher circulates and prompts with 'which side is the hypotenuse of your right triangle?'		
25-40	Class tackles Figure it Out Q7, section 2.7, p.53 ("Construct a square whose area is equal to the difference of the areas of squares of sidelengths 5 and 7 units") together, drawing on the chapter's construction arc. Teacher closes with a full-chapter summary: doubling, halving, the theorem, triples, open problems, and applications.		

**Teacher Notes**

The chapter's closing period should feel like a panoramic review - each problem today echoes a theme from earlier sections. For Q9, students may not immediately see that the altitude creates a right triangle; prompt them to draw it rather than telling them. Q7 is a satisfying final construction problem because it reverses the area-addition idea of section 2.4 into area subtraction.